

LOW-RANK NETWORKS LEARN HIGH-FREQUENCY FEATURES

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ABSTRACT

We study how low-rank random feature (RF-LR) and matrix-rank (MMNN) networks learn representations through *channel feature learning*: in our setting, different low-rank channels specialize to distinct spatial locations and progressively capture higher-frequency components. We combine a tractable mean-field toy model with a rigorous *log-ratio criterion* that characterizes when one channel dominates at a given input location, and we validate the predictions with controlled experiments. We then report an *empirical study of oscillatory complexity across layers*: for MMNNs trained on 1D regression, we define the mean number of strictly-positive local minima per component of the layer-wise partial functions and show that this statistic increases with layer index, consistent with deeper layers learning more oscillatory representations. The pipeline is reproducible; scope is limited to the RF-LR/MMNN setting and 1D regression. Results align with the SciForDL focus on empirical studies of learned representations and training dynamics.

1 INTRODUCTION

Understanding how deep networks learn representations remains a central challenge. The Science of Deep Learning workshop promotes controlled experiments and tractable models to test precise hypotheses. In that spirit, we focus on *feature learning* in low-rank neural networks (RF-LR with frozen random features, and MMNNs): how the r low-rank channels organize representations across space and frequency.

Low-rank architectures factorize weights and reduce parameters while preserving expressivity. In our setting, different channels specialize: each channel tends to dominate near a *distinct spatial value* (localization), and channels separate by *frequency* (lower first, then higher). We summarize the feature-learning and *log-ratio* analysis from the mean-field treatment of RF-LR networks, then present a *controlled empirical study* of oscillatory complexity in MMNNs using the number of strict local minima per partial-function component as a proxy for how oscillatory a layer’s representation is. **Scope:** the theory applies to the RF-LR toy setting; the oscillatory-complexity results are for 1D regression and the reported architecture; we do not claim generalization beyond this setting without further experiments.

2 SETUP: RF-LR AND CHANNEL PARTIAL FUNCTIONS

We consider a low-rank random feature network. The first layer uses frozen random features $L^0(C_1)$; the second layer has a frozen mixing matrix L of rank r and trainable weights $w_1(t, C_1, k)$, $w_2(t, C_2)$. The second-layer pre-activation is $H_2(t, c_2; X, W) = \sum_{k=1}^r L_{c_2, k} f_k(t; X, W)$, where the *channel partial functions* are $f_k(t; X, W) = \mathbb{E}_{C_1}[w_1(t, C_1, k) \varphi_1(\langle L^0(C_1) \rangle X)]$. The backpropagated signal is $B_k(t; X, W) = \mathbb{E}_{C_2}[L_{C_2, k} \varphi'_2(H_2(t, C_2; X, W)) w_2(t, C_2)]$. The kernel induced by the frozen first layer is $K_{\mu_0}(x, x')$ (NNGP-type). The mean-field evolution for f_k is

$$\partial_t f_k(t, x) = -\xi_1(t) \mathbb{E}_{Z=(X, Y)}[d_L(Z; W(t)) B_k(t; X) K_{\mu_0}(x, X)]. \quad (1)$$

3 MECHANISM OF SPIKE LEARNING

When channel k dominates at x_0 , $H_2 \approx L_{c2,k} f_k$ there, so $\varphi'_2(H_2)$ is effectively driven by that channel. The backprop signal B_k aggregates over the second layer; once a channel dominates, the dynamics create *emergent* sign-coherence and a self-reinforcing loop: larger dominance $\rightarrow$ larger $B_k \rightarrow$ larger $\partial_t f_k \rightarrow$ further amplified dominance. The log-ratio criterion (Theorem 4.2) formalizes that *when* certain conditions hold on an interval I , dominance cannot be lost on I .

4 TWO-SIDED STEP TOY MODEL AND LOG-RATIO CRITERION

Consider the *two-sided step*: two support points $x_0 = -\delta$, $x_1 = +\delta$ with $y(x_0) = +A$, $y(x_1) = -A$. For this finitely supported data, f_k takes the form $f_k(t, x) = f_k(0, x) - \Gamma_{k,1}(t) K_{\mu_0}(x, x_0) - \Gamma_{k,2}(t) K_{\mu_0}(x, x_1)$, with $\Gamma_{k,p}(t) = \int_0^t \xi_1(s) d_p(s) B_{k,p}(s) ds$. The full evolution at x_p is $\partial_t f_k(t, x_p) = -\xi_1(t) K_{\mu_0}(x_p, x_p) d_p(t) B_{k,p}(t) + E_p(t)$, with $|E_p(t)| \leq C' \psi(\delta)$, $\psi(\delta) \rightarrow 0$ as $\delta \rightarrow \infty$.

Definition 4.1 (Log-ratio criterion). At x_0 , define $R_{12}(t, x_0) = \log \frac{|f_1(t, x_0)|}{|f_2(t, x_0)|}$.

Theorem 4.2 (Log-ratio growth at x_0 (conditional)). **Setting:** *Two-sided step*, $r = 2$ channels, 3-layer RF-LR.

Hypothesis (at x_0 , for all $t \in I$): (i) $-d_0(t) \geq 0$; (ii) $B_{1,1}(t)$ has the same sign as $f_1(t, x_0)$; (iii) $|B_{2,1}(t)| \leq \rho_0 \frac{|f_2(t, x_0)|}{|f_1(t, x_0)|} |B_{1,1}(t)|$ for some $\rho_0 \in [0, 1)$.

Conclusion: For $t \in I$, $\partial_t R_{12}(t, x_0) = (1 - \rho_0) \xi_1(t) K_{\mu_0}(x_0, x_0) (-d_0(t)) \frac{|B_{1,1}(t)|}{|f_1(t, x_0)|} + \varepsilon_0(t)$, with $|\varepsilon_0(t)| \leq C'' \psi(\delta)$. Hence $\partial_t R_{12}(t, x_0) \geq 0$ whenever the leading term dominates $|\varepsilon_0|$. Strict dominance of channel 1 at x_0 cannot be lost on I . The result is symmetrical at x_1 .

Remark 4.3. The theorem is conditional: it states that *when* (i)–(iii) hold, the conclusion follows. In practice, (i)–(iii) are observed under standard initialization.

5 EXPERIMENTAL VALIDATION: LOG-RATIO DIAGNOSTICS

We validate the log-ratio on frequency-dependent function approximation. Setup: 3-layer low-rank ReLU, $n = 1024$, $r = 15$, target $f(x) = \cos(8\pi x)$ on $[-1, 1]$, $N = 5000$ samples. At $x = 0$ we measure channel partials f_k and log-ratios $R_{i,j} = \log |f_i| - \log |f_j|$. We observe that the trajectory of $\max_{i,j} R_{i,j}(x = 0)$ shows sustained growth (one channel increasingly dominates), and the heatmap of $R_{i,j}$ over training exhibits the same specialization pattern. Newer log-ratio plots (over training and across locations) are more stable; a **uniform distribution in log space** of the ratios is a useful diagnostic for specialization—when channel dominance is well separated, the log-ratios spread and stabilize. A related phenomenon is **log-ratio collapse**: under some LR/batch regimes the log-ratios can collapse (e.g. all channels similar) or, conversely, diverge globally by end of training; scaling laws with batch size also appear (e.g. at lr 10^{-3} , large batch 128 can perform very well in some configs). Figures ?? and ?? illustrate the contrast: at lr 2×10^{-3} , bs 4, trajectories diverge by end of training; at lr 2×10^{-4} , bs 4, they converge (stable specialization).

6 EMPIRICAL ANALYSIS OF OSCILLATORY COMPLEXITY ACROSS LAYERS

A natural hypothesis is that *deeper layers* build increasingly complex, high-frequency internal representations. Testing this in a controlled way requires *observable proxies* for “oscillatory complexity” computable from trained models. We focus on **matrix-rank neural networks (MMNNs)** trained on 1D regression. For each layer we have **partial functions** $h_\ell(x) \in \mathbb{R}^{d_\ell}$ (one scalar per channel) on $x \in [-1, 1]$. A simple, interpretable proxy for how oscillatory a scalar function is along x is the **number of strict local minima** above a positivity threshold $\varepsilon = 10^{-4}$ (to ignore ReLU zero plateaus).

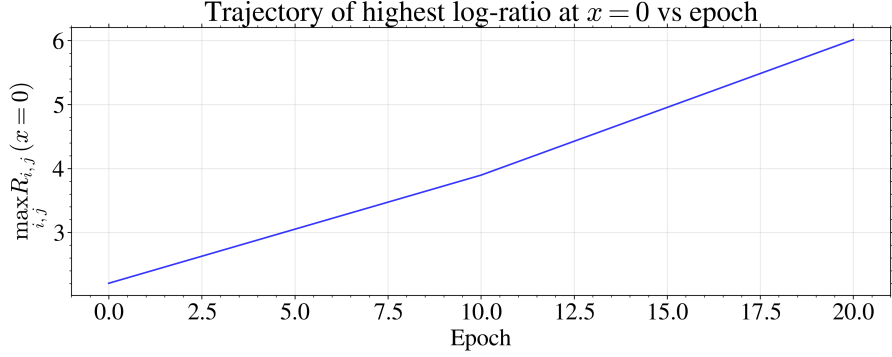


Figure 1: Trajectory of $\max_{i,j} R_{i,j}(x=0)$ vs. epoch. Setup: $n = 1024$, $r = 15$, $\cos(8\pi x)$.

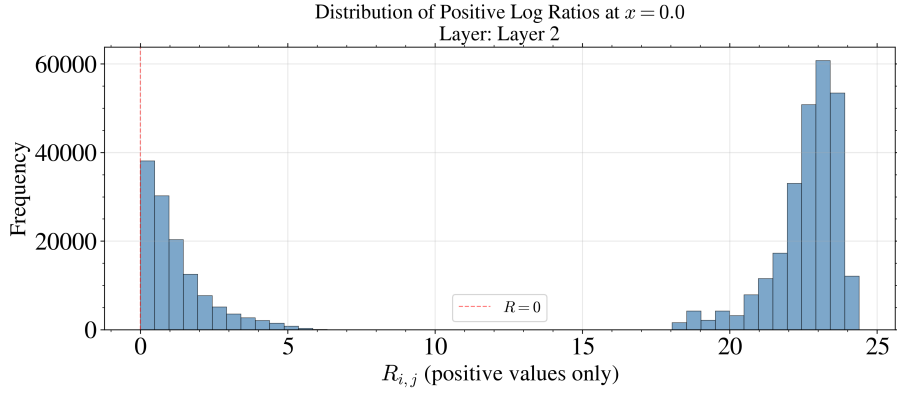


Figure 2: Log-ratios $R_{i,j}$ at $x = 0$ over training (all channel pairs).

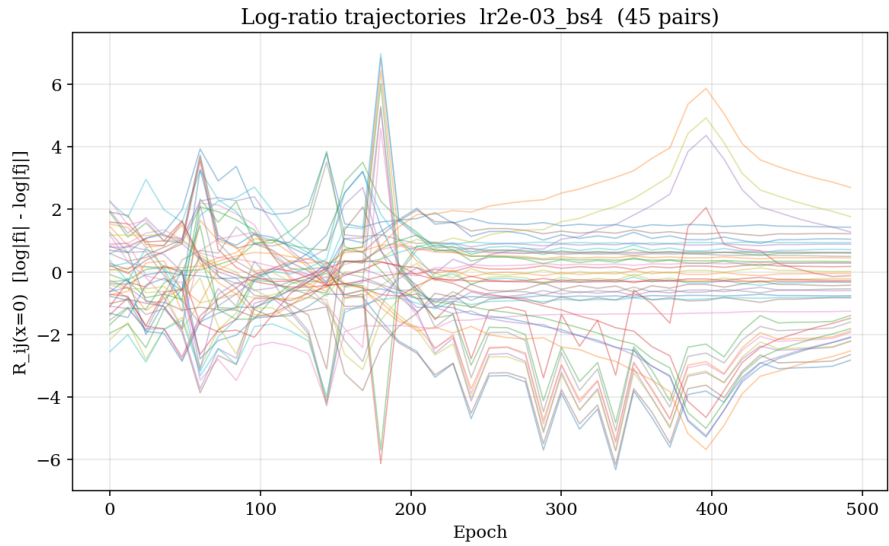


Figure 3: Log-ratio trajectories; $\text{lr } 2 \times 10^{-3}$, bs 4. Divergence by end of training.

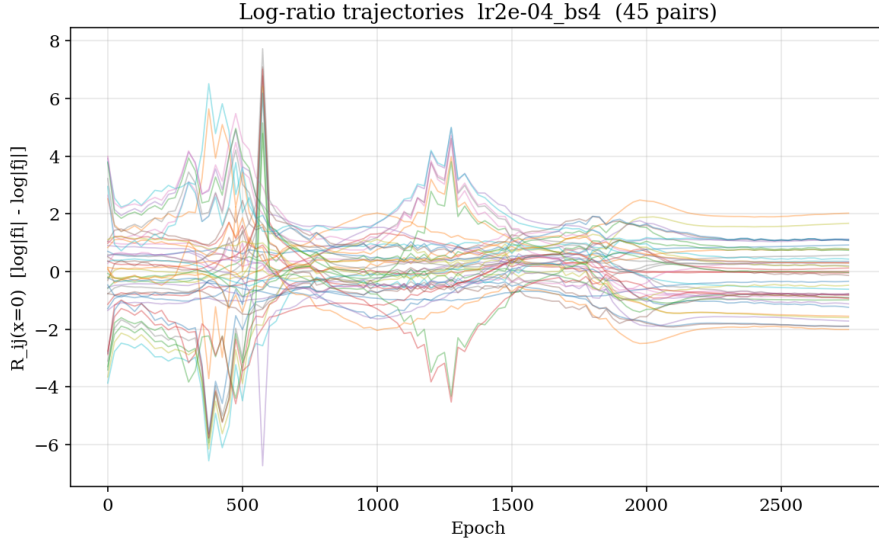


Figure 4: Log-ratio trajectories; $\text{lr } 2 \times 10^{-4}$, $\text{bs } 4$. Convergence (stable specialization).

6.1 STATISTIC AND PROTOCOL

For each component $h_{\ell,j}(x_i)$ on a uniform grid $x_1, \dots, x_N$ ($N = 1000$), we count a *strict local minimum* at index i if $h_{\ell,j}(x_i) < h_{\ell,j}(x_{i-1})$, $h_{\ell,j}(x_i) < h_{\ell,j}(x_{i+1})$, and $h_{\ell,j}(x_i) > \varepsilon$. The per-component count is $m_{\ell,j}$. The **layer-level statistic** is the mean number of minima per component:

$$\bar{m}_\ell = \frac{1}{d_\ell} \sum_{j=1}^{d_\ell} m_{\ell,j}. \quad (2)$$

We also record boxplots and histograms across components. Models are trained on a 1D target (e.g. sum of cosines) with MSE; we use saved checkpoints and the same partial-function extraction as in the repository’s leap analysis.

6.2 RESULTS

In all runs we observe a **clear increase in $\bar{m}_\ell$ with layer index ℓ** in the second half of the network. For $L = 6$, $W = 128$, and $R \in \{5, 10, 15, 20, 30, 36, 50\}$ (one run per R), early layers (1–4) have mean $\bar{m}_\ell \leq 0.35$ for all groups; later layers (roughly 7–13) show mean counts from about 1 to 9, with the highest values at the last representation layer (e.g. layer 12: $\bar{m}_{12}$ ranges from 2.33 for $R = 15$ to 8.60 for $R = 5$; layer 14, scalar output, has a single count between 2 and 11 depending on R). Distributions across components are **right-skewed** and heterogeneous (e.g. at layer 12, quartiles differ strongly across R). This is consistent with the hypothesis that **deeper layers learn more oscillatory partial functions** under this discrete proxy. The dependence on rank R is **not monotonic** in the current single-run-per-config data, so replication over seeds is needed to characterize rank effects.

6.3 LIMITATIONS AND FOLLOW-UPS

Limitations: (1) no seed replication—each (L, W, R) currently has one run; (2) grid resolution—the minima count is a lower bound, so faster oscillations may be under-counted; (3) the last layer is scalar in our setup; (4) conclusions are restricted to the specific 1D regression task and the L, W, R range used. Suggested follow-ups: multiple seeds per config with error bars; optionally a Fourier-based metric (e.g. spectral centroid); extension to other depths and widths.

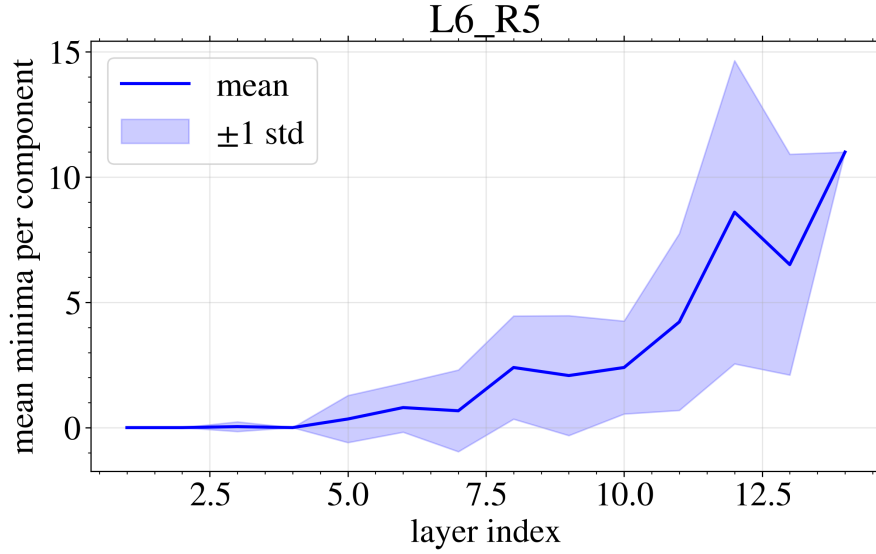


Figure 5: Mean number of strictly-positive local minima per component vs. layer index ℓ , for MMNN with $L = 6$, $W = 128$, $R = 5$. Error bands show ± 1 standard deviation across components.

7 DISCUSSION

The log-ratio $R_{ij}(t, x)$ is a simple diagnostic for channel-wise feature learning; the conditional theorem and two-sided step toy model specify when dominance is maintained. Mean-field two-step and cosine experiments (channel shares at spike locations, log-ratios over time) show increasing channel specialization in agreement with this picture. The oscillatory-complexity study provides empirical evidence that deeper layers exhibit higher mean minima per component (e.g. $\bar{m}_\ell \leq 0.35$ in layers 1–4 vs. 2–9 in late layers at $L = 6$, $W = 128$), consistent with depth supporting higher-frequency structure; rank R modulates the curves without a monotonic trend in the current data. Both strands align with the SciForDL goal of controlled experiments to formulate and test precise hypotheses about deep learning.

A REPRODUCIBILITY

The partial-function minima analysis can be reproduced with the repository script `positive_local_minima_analysis.py` (see `experiments/partialfunctionanalysis/RESULTS_ANALYSIS.md` for commands and outputs). Aggregated statistics (mean $\bar{m}_\ell$ per layer and per group, quantiles across components) are written to `group_layer_stats.csv` and `group_summary.json` under `output_leap_all_L6.W128_grouped/`; the figure used here is `grouped/L6_R5/mean_minima_per_component_vs_layer.png`.

B SCIENCE OF DL IMPROVEMENT CHALLENGE SUBMISSION

B.1 WHAT MODEL ARE YOU TARGETING?

Low-rank neural networks (RF-LR and MMNNs) for 1D regression and scientific-computing-style targets (hierarchical Fourier, sum of cosines). We target understanding of their feature learning (channel specialization, log-ratio dynamics) and oscillatory complexity across layers.

B.2 HOW DO YOUR RESULTS CONTRIBUTE—OR COULD POTENTIALLY CONTRIBUTE—TO UNDERSTANDING THESE MODELS?

We provide a conditional theorem and toy model for when channel dominance is maintained (log-ratio criterion) and an empirical proxy (mean minima per component) showing that deeper layers learn more oscillatory partial functions, with a reproducible pipeline to refine or falsify the claim.

B.3 HOW DO YOU EXPECT YOUR SUBMISSION TO INFLUENCE FUTURE WORK?

The log-ratio diagnostic and the minima-count statistic can be adopted in other architectures and tasks; the oscillatory-complexity protocol can be extended with multiple seeds and Fourier-based metrics to strengthen scaling claims about depth and rank.